# Paper F-02: Galactic Dynamics: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD
Independent Researcher
Founder, Vura Labs
Kampala, Uganda
ORCID: 0009-0001-0556-5516

August 2026

*Shared Part IV notation and evidence boundaries are defined in `PART_IV_METHODS_AND_CLAIM_BOUNDARY.md`. This paper states only its local observables, comparison, and failure conditions.*

**Abstract**

This paper develops a falsifiable CDFD/AFL model of Galactic Dynamics. It maps accretion, star formation, radiation, gas flow, and angular momentum to driving flux $\Phi$, gravity, feedback, cooling, orbital structure, and halo potential to active constraint $C$, spiral waves, bars, outflows, mixing, and orbital reorganization to responsiveness $S$, and mergers, metallicity, stellar populations, and orbital history to structural memory $M_s$. The target outcome is morphology, star formation, rotation, and long-term evolution, tested against multiwavelength surveys, kinematics, gas maps, stellar populations, and simulations. The paper treats universality as a shared model architecture rather than an assertion that unlike systems have the same material mechanism. It states the negative result that would make the mapping unnecessary and places the domain claim inside the cumulative CDFD programme and its reproducible runtime boundary.

## 1 Introduction

A galaxy is an open thermodynamic system. Gas flows inward from the cosmic web, stars form and migrate, and energy is violently expelled outward by supernovae and active galactic nuclei (AGN). The delicate balance of these forces creates stable, long-lived structures such as spiral arms and galactic bars.

In standard ΛCDM cosmology, these structures are shaped by the dominant gravitational well of a dark matter halo. However, under the CDFD framework, dark matter is re-evaluated. Whether it is a distinct particle or a topological property of the vacuum itself (as posited in earlier papers of this series), its functional role in the galaxy is unambiguous: it serves as the master structural constraint ($C$) that binds the immense radial throughput ($\Phi$) of the galactic disk.

## 2 The Galactic Flow Network

We apply the core Adaptive Flux Limitation equation to the galactic scale:

$$\Psi_s = \left(\frac{\Phi}{C}\right) \cdot S \cdot M_s \tag{1}$$

Here, the parameters of the galactic "organism" are:

- **Flux ($\Phi$)**: The kinetic flow of stars and the thermodynamic flux of the interstellar medium (ISM). High star formation rates (SFR) correspond to peaks in $\Phi$.
- **Constraint ($C$)**: The gravitational binding energy. In the outer radii of the galaxy, the constraint $C$ must remain high to prevent the high-velocity stellar flow ($\Phi$) from causing $\Psi_s \gg 1$ (which would result in dissolution of the galactic disk). The "dark matter halo" provides exactly the required geometric constraint.
- **Responsiveness ($S$)**: The density of cold molecular gas capable of collapsing into new stars.
- **Memory ($M_s$)**: Spiral density waves and metallicity gradients. Spiral arms are essentially kinetic "memory banks"—standing wave patterns that remember the gravitational history of passing molecular clouds.

### 2.1 Density Waves as Memory Lattices

Lin-Shu density wave theory describes spiral arms as regions of enhanced density through which stars and gas flow, analogous to a traffic jam. In CDFD, this traffic jam is a chronic constraint applied to the flow. When gas enters the spiral arm, $C$ spikes locally, forcing $\Phi$ to bottleneck. The resulting compression triggers star formation, briefly raising the responsiveness $S$.

If the memory term $M_s$ (the density wave pattern) locks the galaxy into too strict of a constraint, star formation ceases as the gas is entirely consumed or blown away. This perfectly describes the transition from active, blue spiral galaxies (Sustained Growth, $\Psi_s > 1$) to "red and dead" elliptical galaxies (Decay/Recession, $\Psi_s < 1$), where the gas flux has dropped to zero and only the kinetic memory remains.

## 3 Dark Matter as Vacuum Constraint

As observed in galaxy rotation curves, the rotational velocity $v(r)$ remains constant at large radii. Under the CDFD formulation, a constant $v(r)$ implies that the ratio $\Phi/C$ is invariant across the outer disk. As the visible matter density ($S$) drops off exponentially, the system maintains $\Psi_s \approx 1$ only if the constraint term $C$ scales appropriately with radius.

The dark matter halo is therefore the necessary vacuum response to galactic-scale angular momentum. Just as an expanding biological tissue induces the growth of surrounding extra-cellular matrix to support it, the spinning galactic disk induces a topological constraint in the surrounding adaptive vacuum.

# 4 Measurement Layer

# 5 Cross-Domain Model Upgrade: Mechanism, Scale, and Tests

## 5.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Galactic Dynamics, the proposed drive is accretion, star formation, radiation, gas flow, and angular momentum. The active limitation is gravity, feedback, cooling, orbital structure, and halo potential. Responsiveness is carried by spiral waves, bars, outflows, mixing, and orbital reorganization, while retained state is represented by mergers, metallicity, stellar populations, and orbital history. The outcome to explain is morphology, star formation, rotation, and long-term evolution.

Table 1: Operational CDFD/AFL map for Paper F-02.

| Term | Paper-specific observable meaning | Measurement question |
|---|---|---|
| $\Phi$ | accretion, star formation, radiation, gas flow, and angular momentum | What rate, load, gradient, or demand enters the focal system? |
| $C$ | gravity, feedback, cooling, orbital structure, and halo potential | Which measured bottleneck limits transfer, function, or recovery? |
| $S$ | spiral waves, bars, outflows, mixing, and orbital reorganization | Which process changes effective capacity or reroutes the load? |
| $M_s$ | mergers, metallicity, stellar populations, and orbital history | Which retained state makes equal present loads produce different futures? |
| $Y$ | morphology, star formation, rotation, and long-term evolution | Which outcome is predicted out of sample, with uncertainty? |

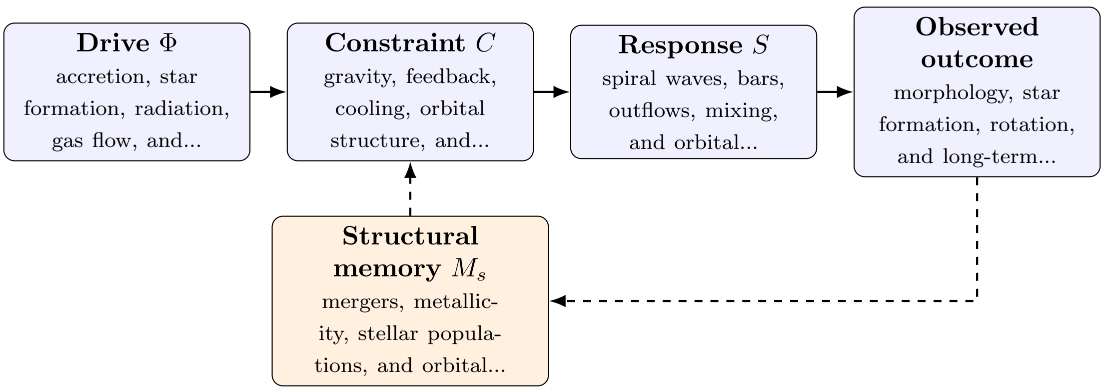


Figure 1: Paper F-02 observable CDFD/AFL architecture for Galactic Dynamics. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

## 5.2 Paper-Specific Causal Hypothesis

For this paper, the hypothesized causal sequence is specific. A change in accretion, star formation, radiation, gas flow, and angular momentum arrives faster than spiral waves, bars, outflows, mixing, and orbital reorganization can compensate. This raises or redistributes gravity, feedback, cooling, orbital structure, and halo potential. If the load persists, the retained state encoded by mergers, metallicity, stellar populations, and orbital history changes the next response, so a later event of equal magnitude can produce a different trajectory in morphology, star formation, rotation, and long-term evolution. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

### 5.3 Cross-Scale Translation Without Mechanistic Erasure

For the cosmic and subatomic systems family, the cross-domain translation must remain subordinate to conservation laws, relativity, quantum mechanics, and established astrophysical or plasma equations. CDFD is useful only if it compresses a regime transition or hysteresis question without pretending that biological adaptation and physical response are the same mechanism. [3, 2, 1] The required scale declaration for this paper is thousands to billions of years; clouds to galaxy clusters. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

### 5.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper F-02. Each rung can fail independently.

| **Test** | **Design** | **Discriminating result** |
|---|---|---|
| Proxy audit | Measure multiwavelength surveys, kinematics, gas maps, stellar populations, and simulations and preregister how each record maps to $\Phi$, $C$, $S$, $M_s$, and $Y$. | The mapping fails if proxies are non-identifiable, circular, or change meaning across cases. |
| Load ramp | Apply or observe a matched comparison across galaxies with contrasting merger or feedback histories while estimating the dominant constraint and response time. | A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint. |
| Recovery and memory | Match present load across units with different histories, then compare recovery paths. | The memory claim requires history-dependent divergence after current conditions are controlled. |
| Model comparison | Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data. | The paper's stated falsifier is: the mapping adds no predictive value beyond standard galactic dynamics. |

### 5.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from multiwavelength surveys, kinematics, gas maps, stellar populations, and simulations and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of morphology, star formation, rotation, and long-term evolution. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the cross-domain translation audit.

## 6 Conclusion

Galactic evolution mirrors biological aging. A galaxy begins with high flux and responsiveness, generating stars aggressively. Over billions of years, it builds structural memory in the form of spiral density waves and supermassive black holes. Eventually, these constraints lock the system, halting the throughput of fresh gas and driving the galaxy into a quiescent state. By analyzing galaxies through the CDFD equation, we unify the astrophysics of dark matter and spiral morphology with the universal principles of adaptive flux limitation.

## Conflict of Interest

The author declares no conflict of interest.

## Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

## 7 Reference Layer

## References


[1] Per Bak, Chao Tang, and Kurt Wiesenfeld. Self-organized criticality: An explanation of the 1/f noise. *Physical Review Letters*, 59(4):381–384, 1987. DOI: `https://doi.org/10.1103/PhysRevLett.59.381`.

[2] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948. DOI: `https://doi.org/10.1002/j.1538-7305.1948.tb01338.x`.

[3] A. M. Turing. The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society of London. Series B*, 237(641):37–72, 1952. DOI: `https://doi.org/10.1098/rstb.1952.0012`.